

Design and Fabrication of fully differential G_m cell, and biquad for Active Filter applications

Prachi Bindal, Hardik Jain

220102071, 220102039

Department of Electronics and Electrical Engineering

Indian Institute of Technology, Guwahati

CONTENTS

I	Gm-Cell Design	3
I-A	Design of Inverter	3
II	Layout of Gm-Cell	4
III	Post Layout Simualtion of Gm-Cell	6
IV	Biquad made from Inverter based G_m cell	6
V	Layout of Biquad	8
VI	Post Layout Simulation of three cascaded Biquads	9
VII	Testing and Test Buffer circuit	10
VIII	Inverter chain Test Circuit	10
IX	Layout of the Inverter Chain test buffer	12
X	Design of Common Drain	12
XI	Layout of Common Drain	14

XII	Stability of Buffer	16
XIII	Combining layout of filter with test buffer	17
XIV	Decoupling Capacitors	17
XV	Final Layout of Biquad with Test Circuits and Pins	18
XVI	Final Post layout simulation with the test buffers across corners	18

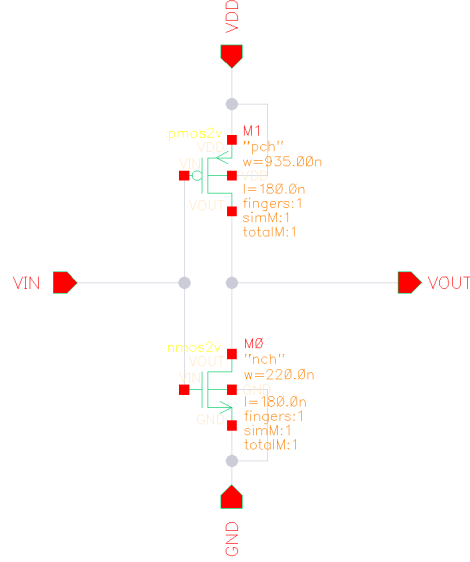


Fig. 2: Inverter Design Parameters

The results were verified by plotting the Bode gain and phase plots. The bandwidth was found to be approximately 400 MHz, and the DC gain was measured to be 17.2 dB.

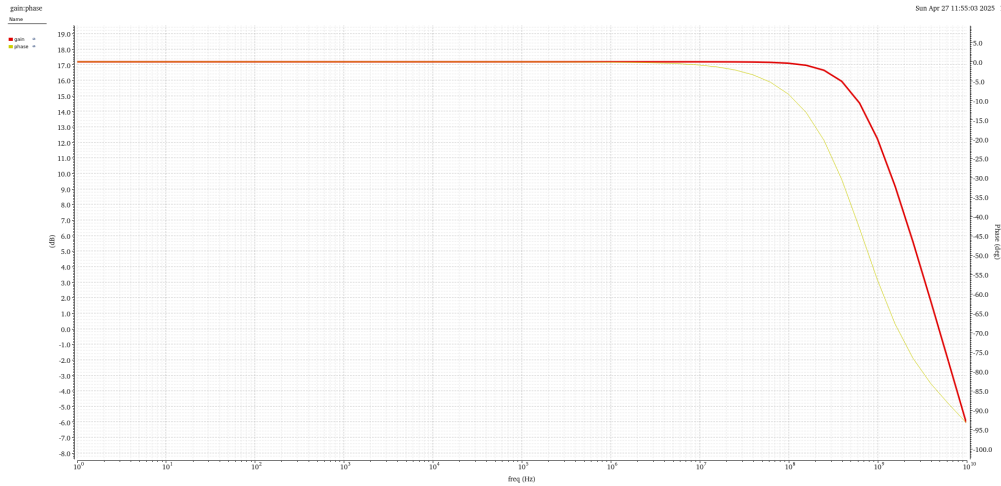
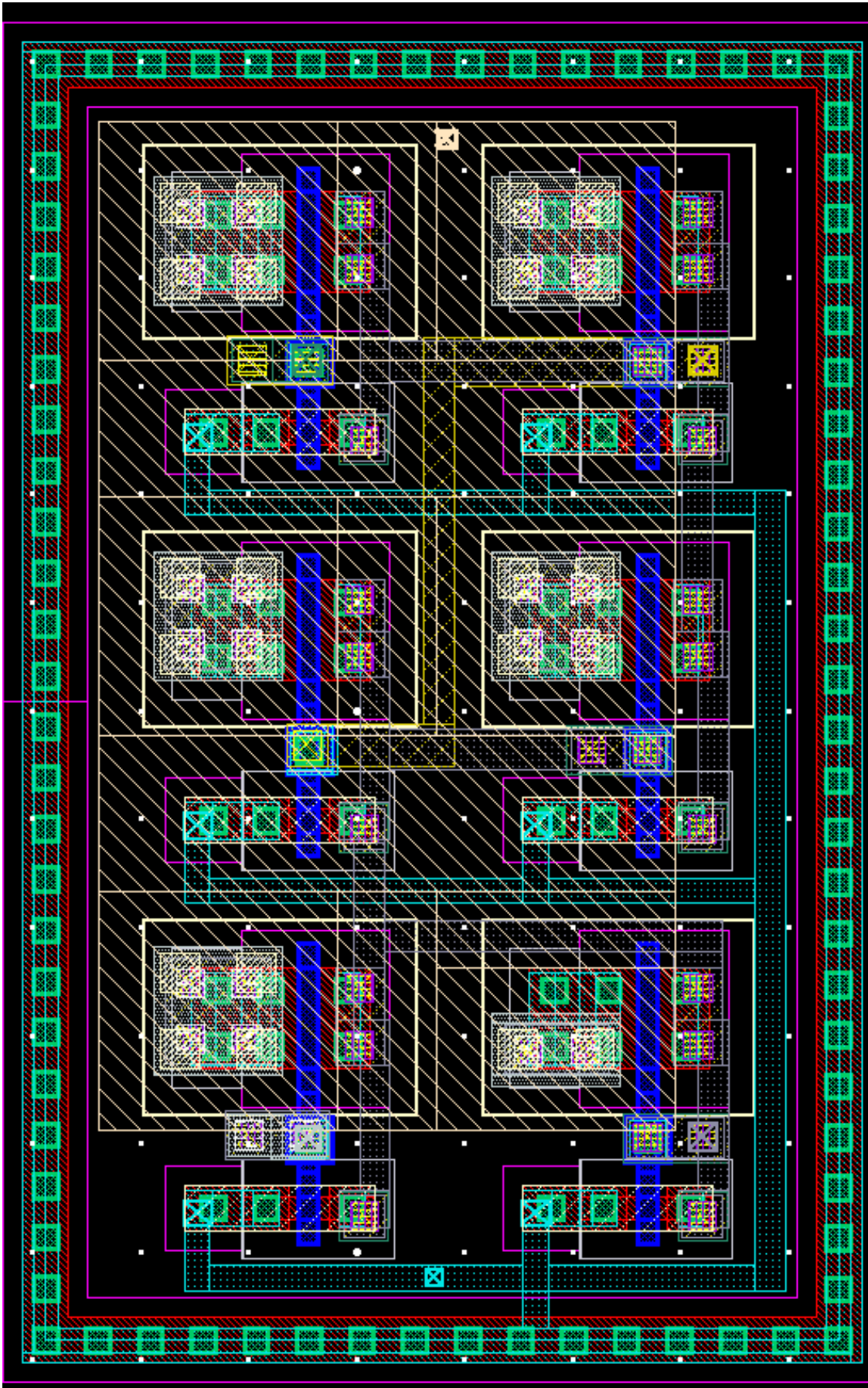


Fig. 3: Bode Plot of Gm Cell

II. LAYOUT OF GM-CELL

A P-sub guard ring was added to ensure sufficient ground contacts. Adequate vias were placed to minimize parasitic resistance. Additionally, the inverters were arranged symmetrically to optimize the differential Gm-cell design.



III. POST LAYOUT SIMULATION OF GM-CELL

Post-layout simulation of the Gm-cell was performed to verify whether parasitic capacitance affected the cutoff frequency.

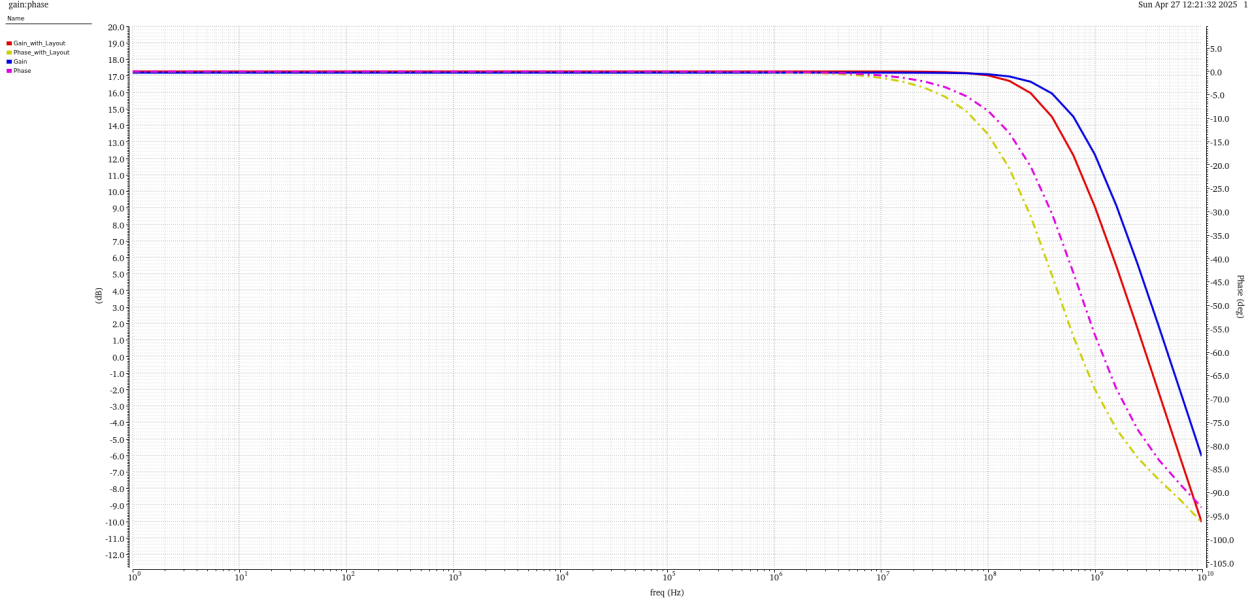


Fig. 5: Gm-Cell Post Layout Bode Plot

As shown in the Bode plot, the cutoff frequency shifted slightly, but this effect is considered negligible for the use of the Gm-cell in the biquad design.

IV. BIQUAD MADE FROM INVERTER BASED G_m CELL

A fully differential biquad is designed with this G_m cell on TSMC 180nm technology. The integrating capacitors are placed symmetrically so that they remain open for common-mode and operational for differential mode.

The bode gain and phase plot of the biquad was plotted. The roll-off is expected to be 40 dB per decade and the phase is expected to fall by 180° over the span of two decades, due to presence of a pair of complex conjugate poles, with cutoff frequency $\omega_p = G_m/C$.

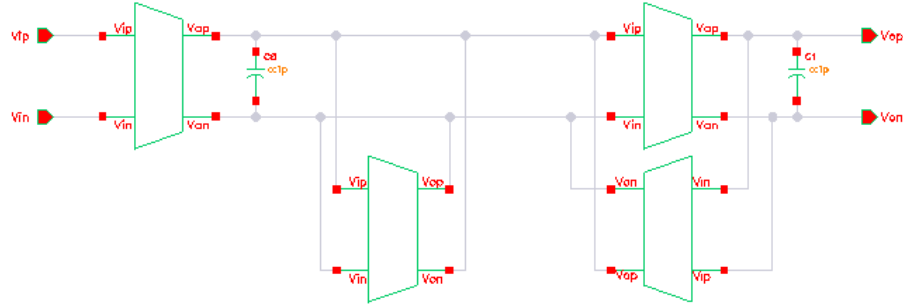


Fig. 6: Biquad made with inverter based G_m cells

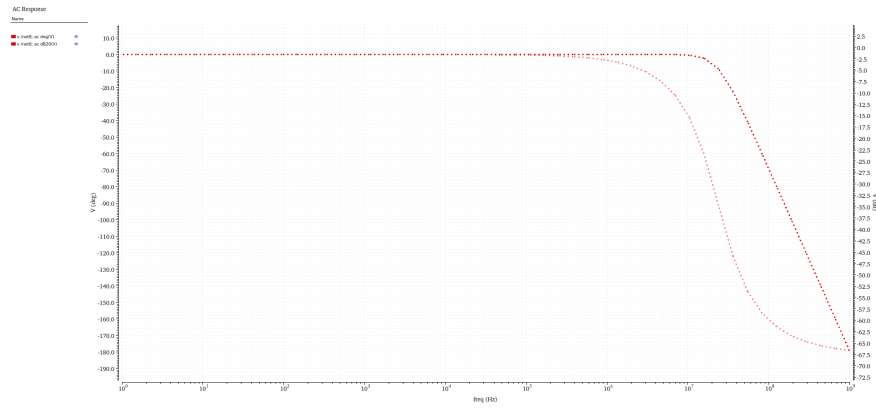


Fig. 7: Bode Plot of the biquad made with inverter based G_m cells

For a higher roll-off, a 6th order filter was designed by cascading 3 such biquads.

The cutoff frequency was observed to be close to 10MHz for integrating capacitor of 1pF .

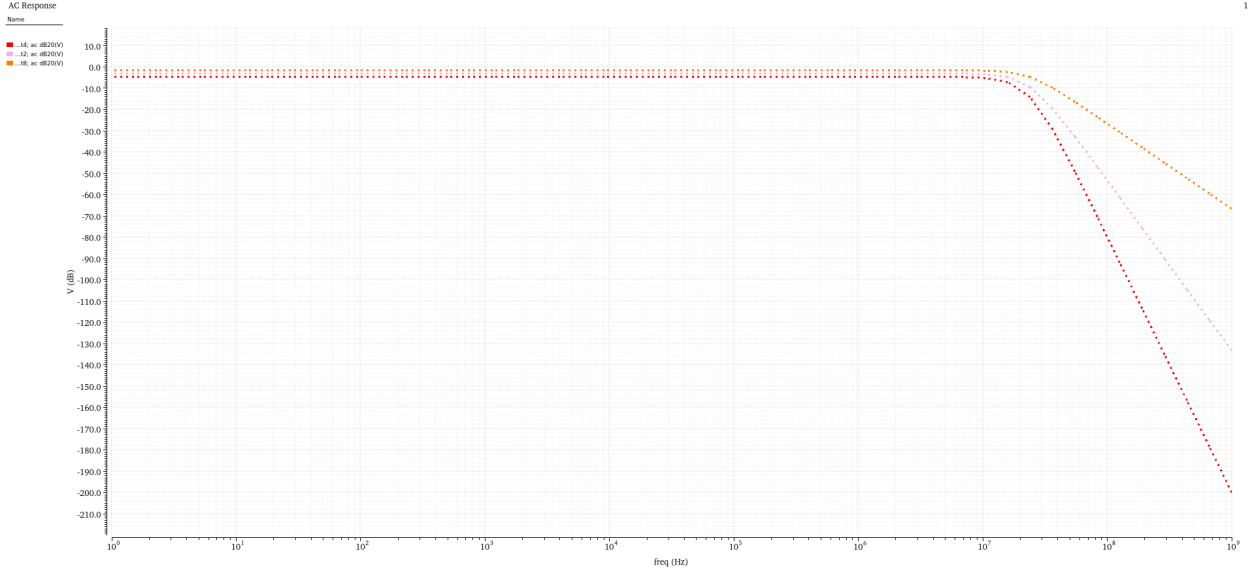


Fig. 8: Bode Plot of the 6th order LPF with integrating capacitor of 1 pF

V. LAYOUT OF BIQUAD

The layout for the biquad was done. The integrating capacitors were matched to reduce process variations. They were also broken into small unit capacitors for better matching.

The layout of 3 biquads cascaded was done, by cascading the layouts of individual biquad.

V_{dd} was taken at the highest metal (Metal 6), ground was taken at the lowest metal (Metal 1), and other routings were done on Metal 2, 3 and 4. This norm was followed throughout the layout procedure.

Sufficient ground contacts are provided since each G_m cell has its own p-sub guard ring. G_m cells are surrounded by dummies.

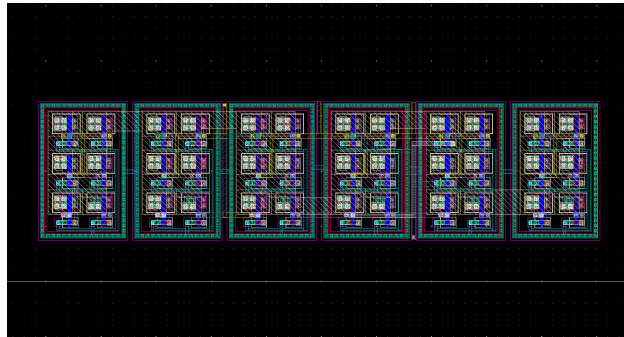


Fig. 9: G_m cells combined

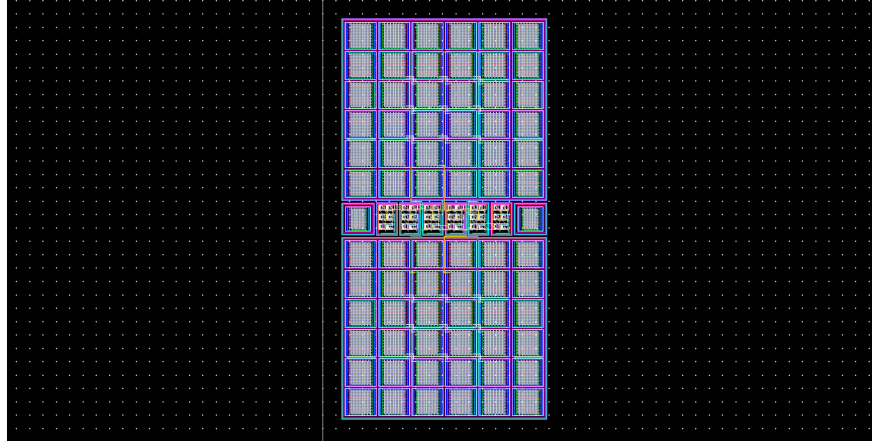


Fig. 10: Layout of Biquad

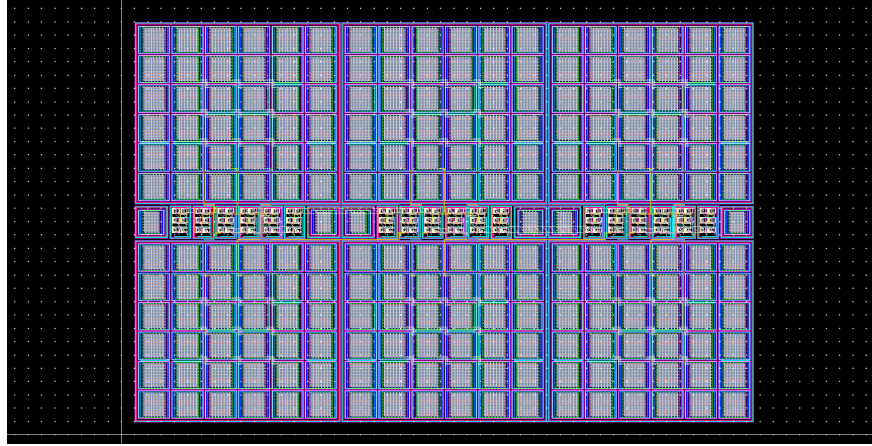


Fig. 11: Layout of three cascaded biquads

VI. POST LAYOUT SIMULATION OF THREE CASCADED BIQUADS

Due to parasitic capacitances, the cutoff frequency reduced to $4MHz$, and to get it back to the designed value of $10MHz$. It was found out that the dominating parasitics are not from the G_m cells, they were coming from the capacitor. So, the integrating capacitors were reduced by the amount of parasitics so that they add to the integrating capacitor and restore the original integrating capacitor, thus restoring the original pre-layout cutoff frequency of $10MHz$. The value of the integrating capacitors were reduced to $0.75 pF$.

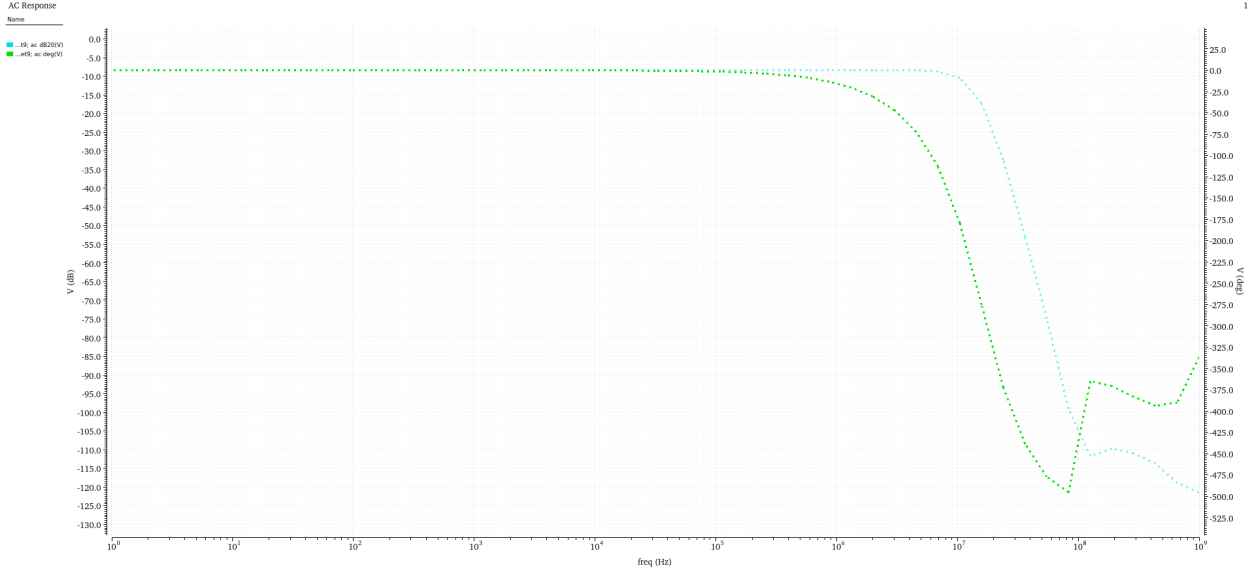


Fig. 12: Post Layout Simulation of three cascaded Biquads

VII. TESTING AND TEST BUFFER CIRCUIT

Since the integrating capacitors are of $1pF$, the oscilloscope probe used for testing would load the filter, since it offers an impedance of $12pF$ in parallel with $1M$ ohms. To mitigate this loading effect, two dedicated test circuit was developed.

VIII. INVERTER CHAIN TEST CIRCUIT

So, a chain of inverter test buffer was designed, in which the first stage acts as an amplifier, to push the filter output above the trip point of the inverter, so that the next inverter pushes it to rails. Thus, we obtain a rail to rail square wave output with the same fundamental frequency as the input.

For better driving of the load, every inverter is sized twice than its preceding inverter. We used three stages since we had to drive a $12pF$ capacitive load with decent rise and fall times.

Now the circuit was tested with the impedance offered by the oscilloscope probe loading effect along with the test buffer. The inverter's output resistance along with the probe's capacitance offers a pole at roughly $750kHz$, the effect of which can be calibrated during testing. The filter response after this is preserved. The AC and transient responses were plotted.

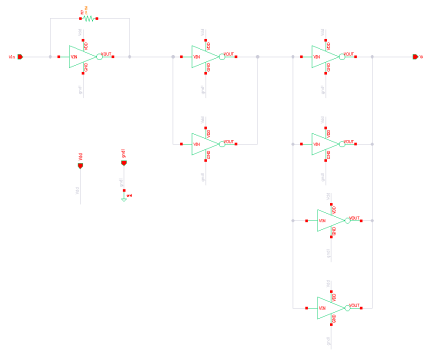


Fig. 13: Chain of inverter buffer

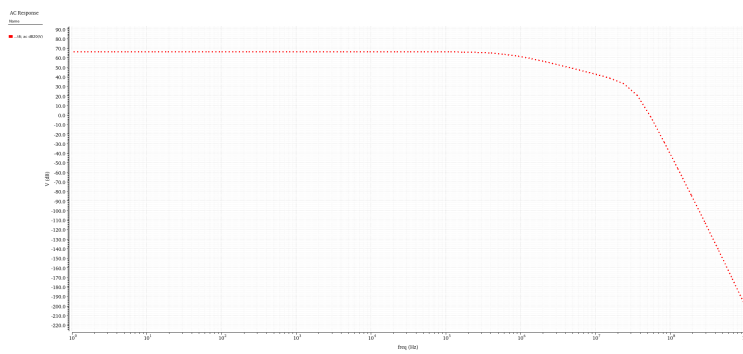


Fig. 14: AC Response of the filter with inverter buffer

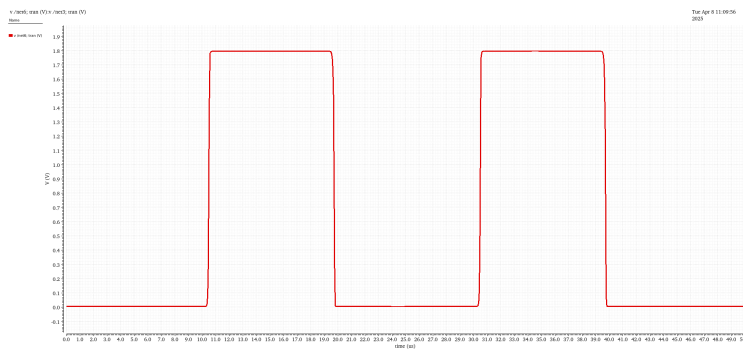


Fig. 15: Inverter based test buffer passband output

IX. LAYOUT OF THE INVERTER CHAIN TEST BUFFER

The layout of the chain of inverter was done. We used rpplus resistor which is a 3 terminal resistor. We used a n-well guard ring for the bulk of the resistor. A serpentine shape was used for the layout of the resistor. Each inverter is surrounded by a p-sub guard ring for sufficient ground contacts.

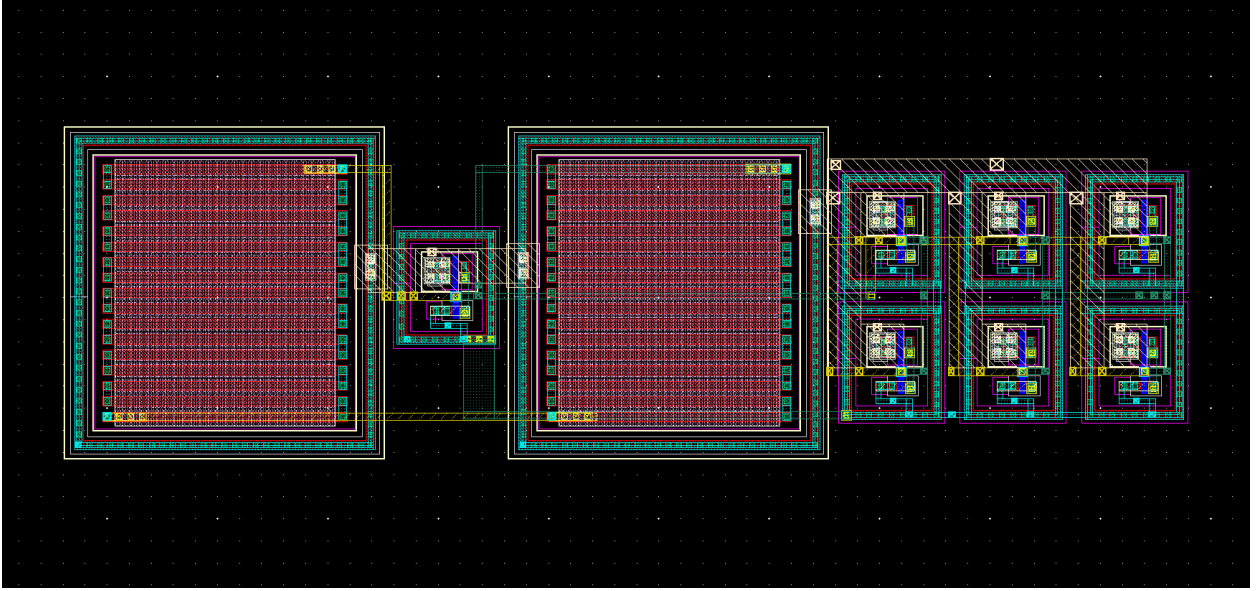


Fig. 16: Chain of inverter layout

X. DESIGN OF COMMON DRAIN

A buffer was selected as the most appropriate solution, and a simple common-drain (source follower) buffer was designed for this purpose. The cutoff frequency of the buffer is given by:

$$f_{\text{cutoff}} = \frac{g_m}{2\pi C_L} \quad (1)$$

To prevent the buffer from affecting the biquad's performance, it was ensured that:

$$\frac{g_m}{2\pi C_L} \gg 10 \text{ MHz} \quad (2)$$

The bias point was set at 900 mV. Due to the body effect in the upper NMOS transistor, its threshold voltage increased to approximately 590 mV. Assuming an overdrive voltage of 180 mV

for the lower transistor, the required current was estimated to be around $500\text{ }\mu\text{A}$. Since a $50\text{ k}\Omega$ resistor had already been laid out, it was chosen for biasing.

To generate the required gate voltage of approximately 720 mV ($540\text{ mV} + 180\text{ mV}$), the aspect ratio (W/L) of the current mirror transistor was swept until the desired voltage was achieved. This branch of the current mirror produced $25\text{ }\mu\text{A}$. Therefore, an aspect ratio 25 times greater was selected for the main current transistor in the buffer. For the upper transistor in the buffer, the W/L ratio was further adjusted to obtain a source voltage of 180 mV .

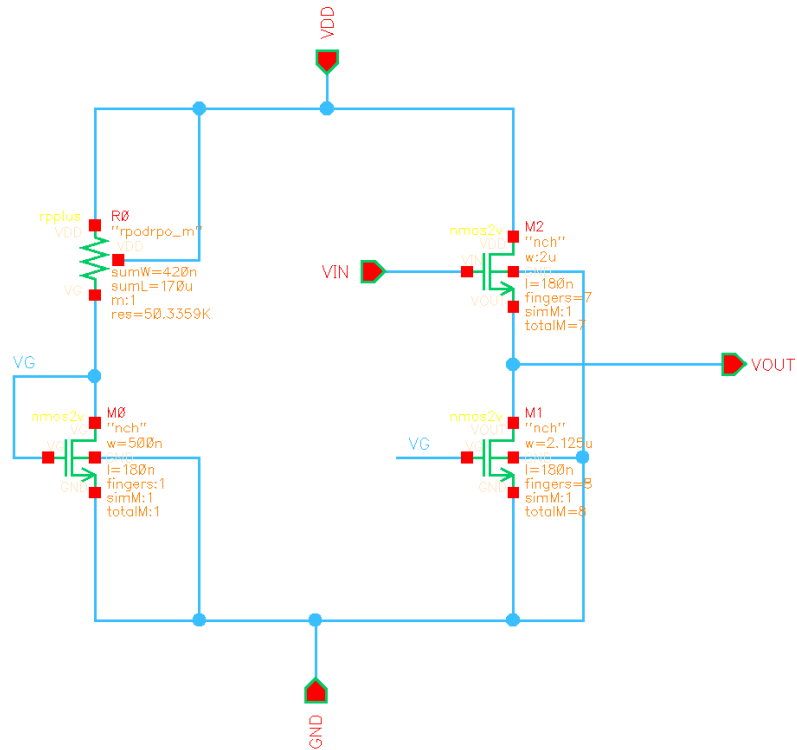


Fig. 17: Common Drain Design Parameters

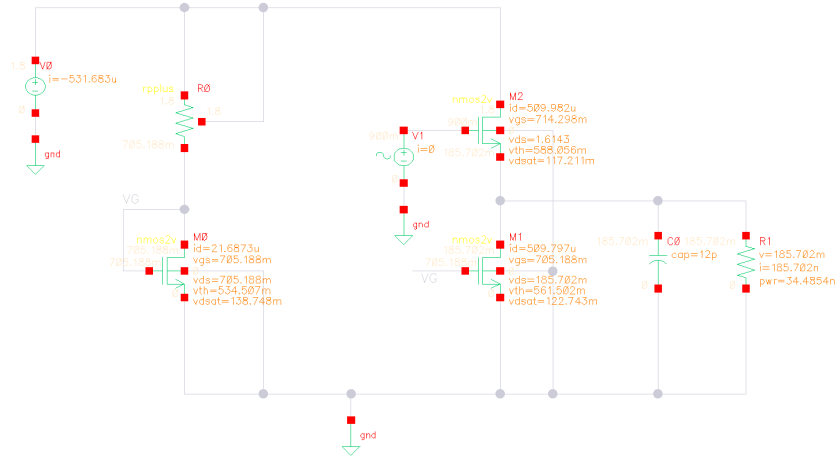


Fig. 18: Common Drain Operating Points

XI. LAYOUT OF COMMON DRAIN

The RPPlus resistor, a three-terminal resistor, was utilized for implementing a $50k\Omega$ resistor in the layout. An n-type guard ring was incorporated around the resistor to ensure proper bulk isolation. The layout of the Drain was carefully designed to align with the overall layout of the biquad filter, maintaining a rectangular shape suitable for compact integration.

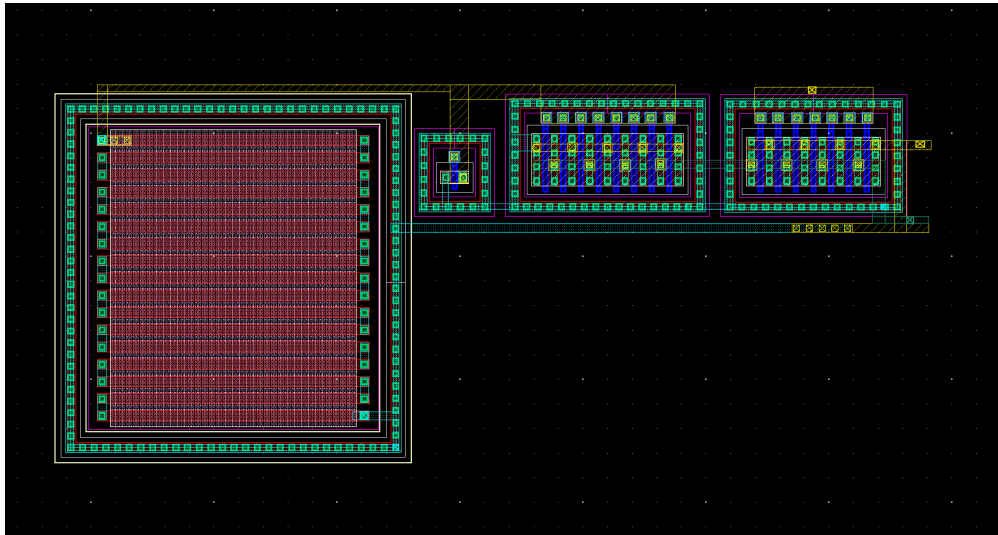


Fig. 19: Layout of Common Drain

AC and transient analyses were performed to verify the functionality of the test circuit,

considering the load impedance comprising a 12 pF capacitor and a 1 M Ω resistor. Additionally, a 10 μ F capacitor was added to ensure that no DC current flows into the load impedance.

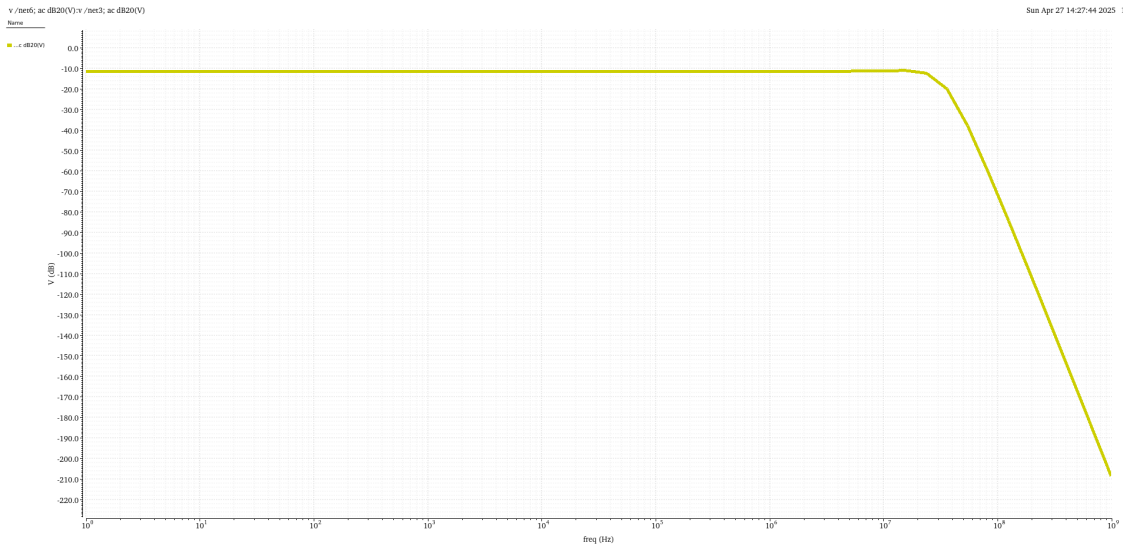


Fig. 20: Bode Plot for Gm-cell with Test circuit

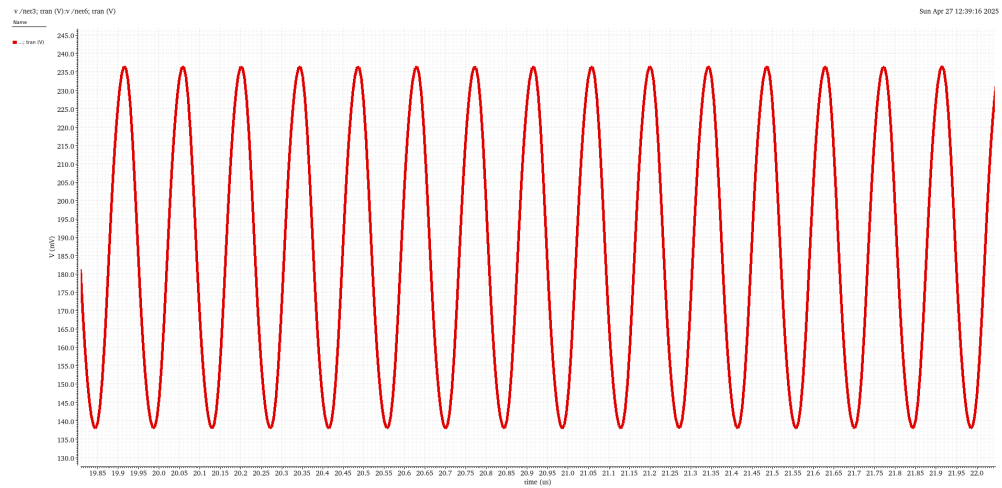


Fig. 21: Tranient Analysis for Gm-cell with Test circuit

From the figure, it can be observed that the probe loading effect was effectively eliminated by using a test buffer.

XII. STABILITY OF BUFFER

Input and Output Impedance of Common Drain buffer were plotted to verify the stability of the Circuit. Input Impedance is not becoming negative at any frequency and also output Impedance is increasing slightly.

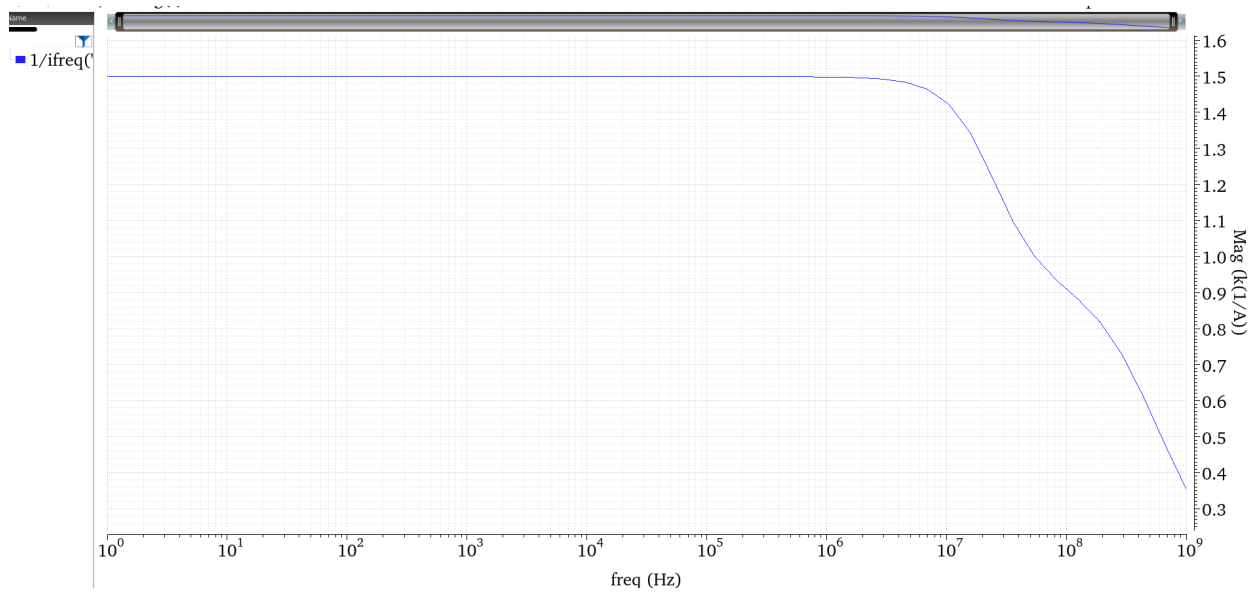


Fig. 22: Input Impedance of Common Drain

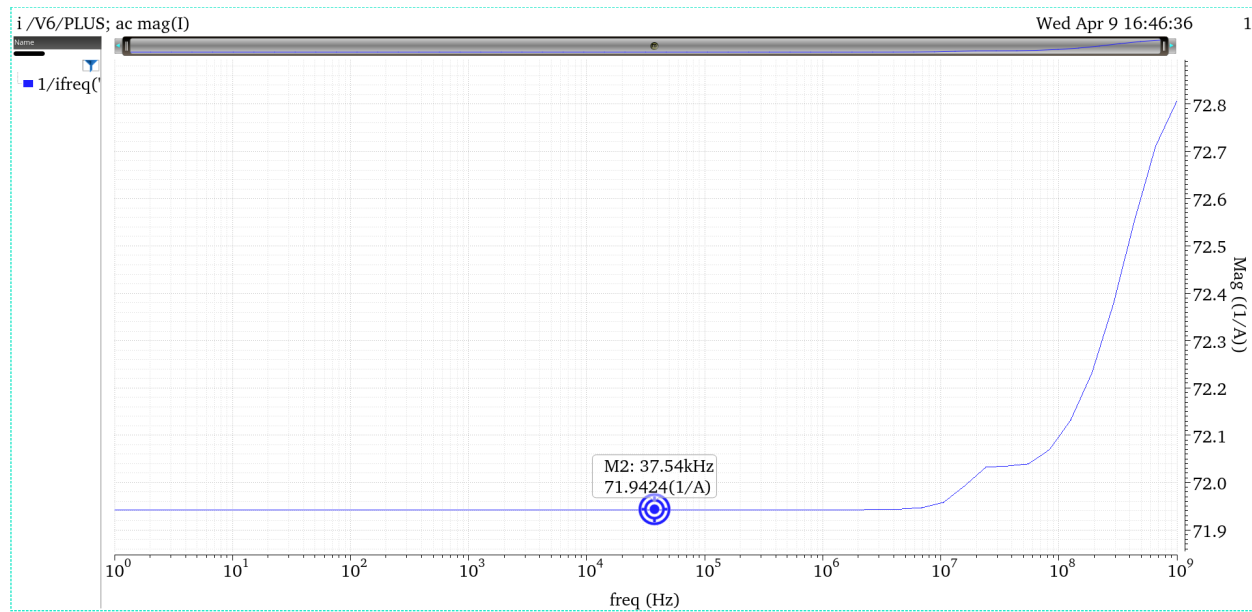


Fig. 23: Output Impedance of Common Drain

XIII. COMBINING LAYOUT OF FILTER WITH TEST BUFFER

The test buffer circuit's layout was combined with the original filter's layout.

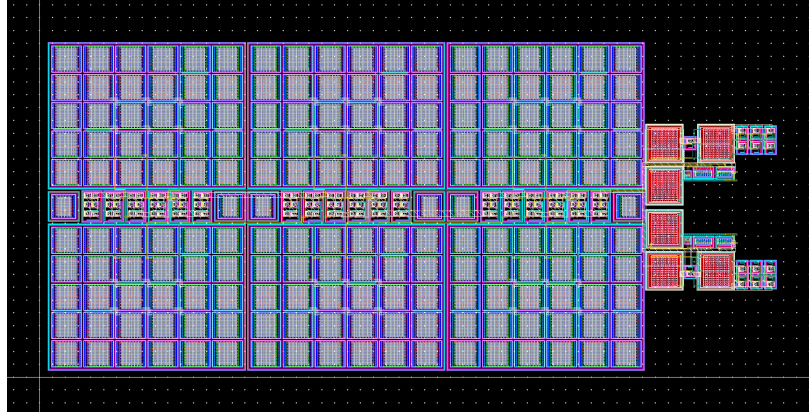


Fig. 24: Combined filter with chain of inverter test buffer

XIV. DECOUPLING CAPACITORS

To supply instantaneous AC current to all circuits locally, and to reduce the quality factor of the supply line, lossy varactor mos capacitors are added between V_{dd} and ground lines.

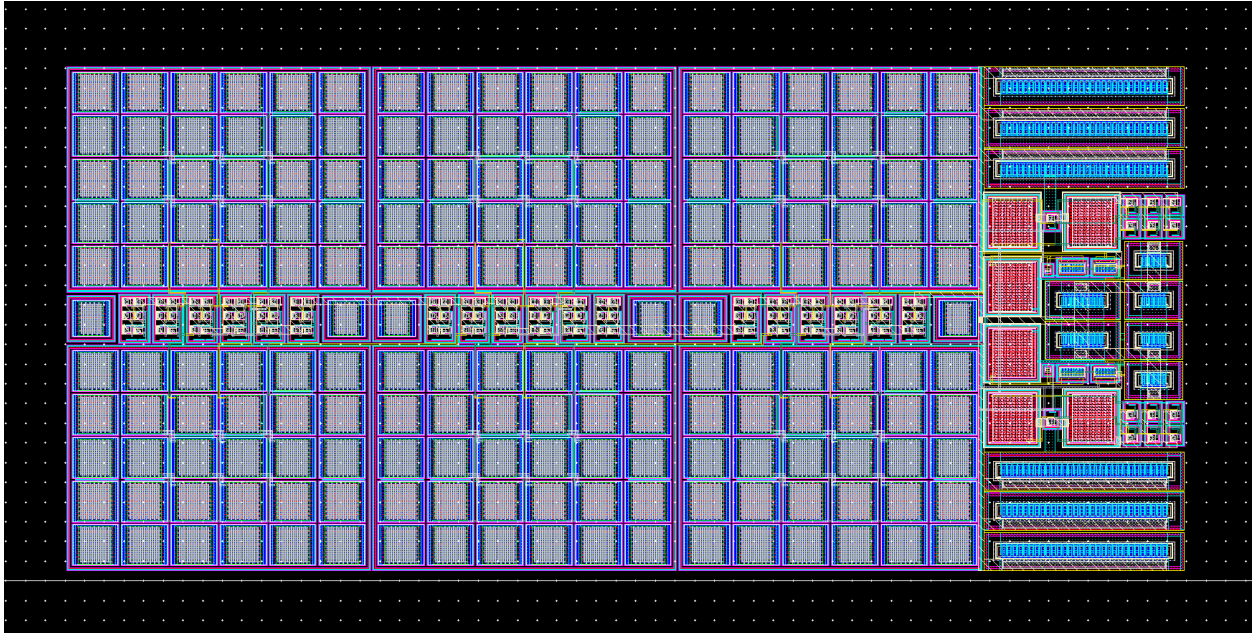


Fig. 25: Final Layout with decoupling capacitors

XV. FINAL LAYOUT OF BIQUAD WITH TEST CIRCUITS AND PINS

For the final tapeout, the pads must be spaced $100\mu\text{m}$ apart from center to center and should each have dimensions of $60\mu\text{m} \times 60\mu\text{m}$. Therefore, appropriate pins were routed to connect to the pads.

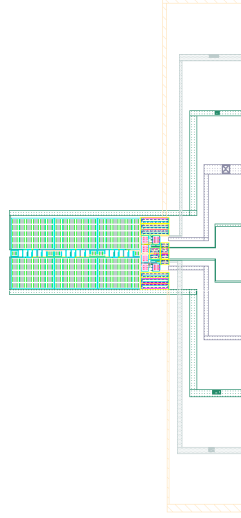


Fig. 26: Final Layout with Spacing for Pads

XVI. FINAL POST LAYOUT SIMULATION WITH THE TEST BUFFERS ACROSS CORNERS

A final post layout simulation was run on the combined circuit and the cutoff frequency was observed to be close to the designed value of 10MHz in the presence of the probe's loading effect. The power consumption is 9.78mW in typical corner, 13.38mW in fast corner and 7.04mW in slow corner.

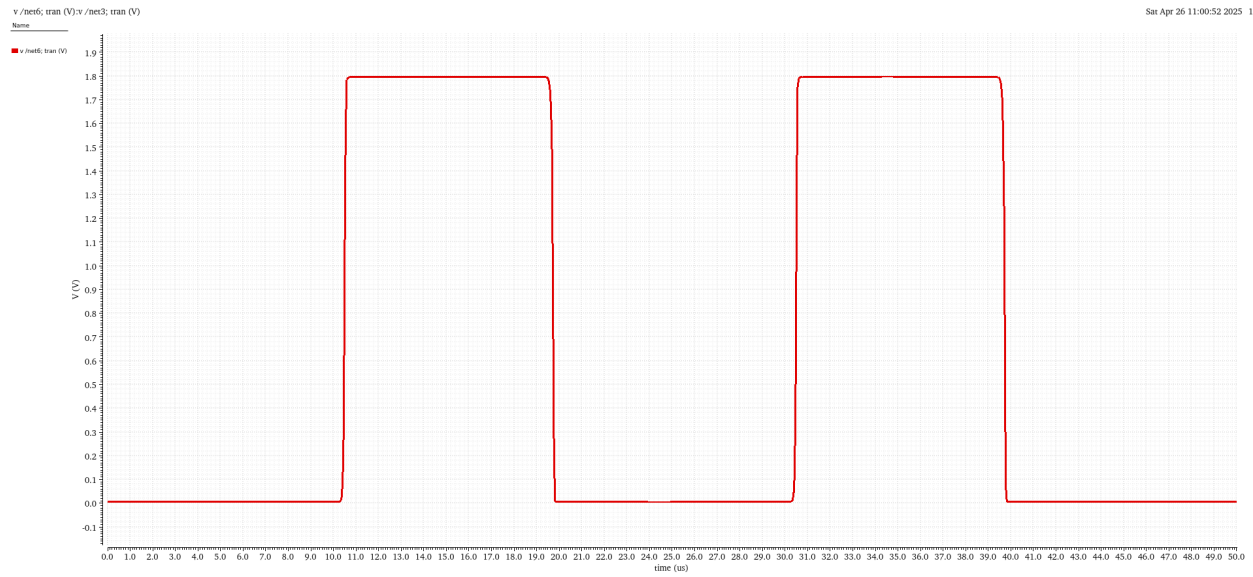


Fig. 27: Post Layout simulation, Passband waveform of inverter test buffer

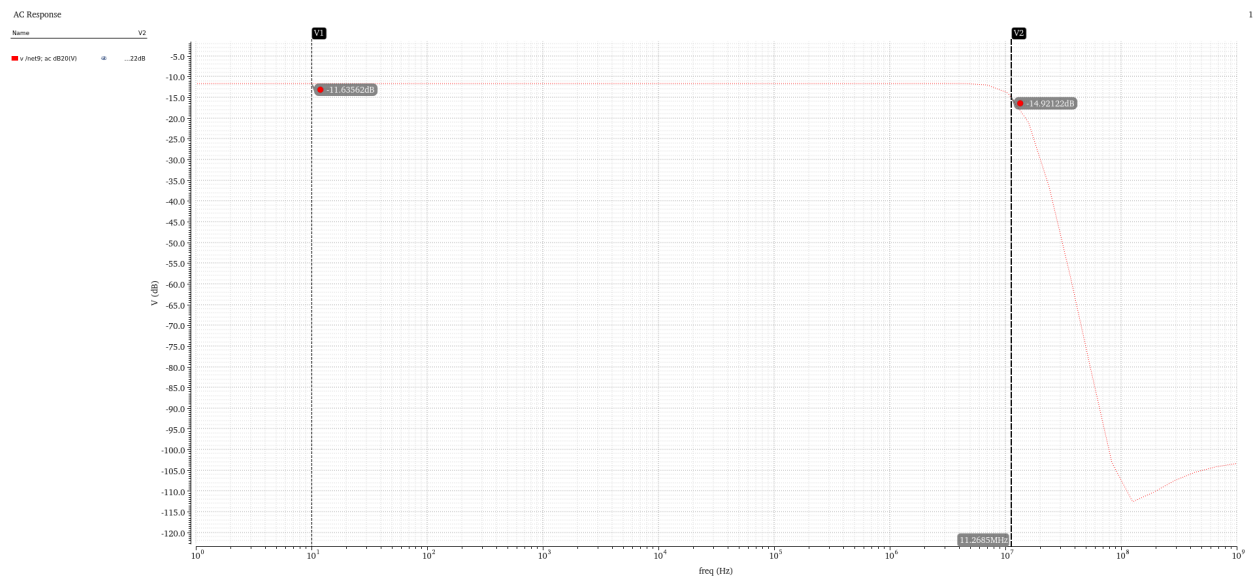


Fig. 28: Post Layout Bode Plot for Gm-cell with Test circuit - Typical Corner

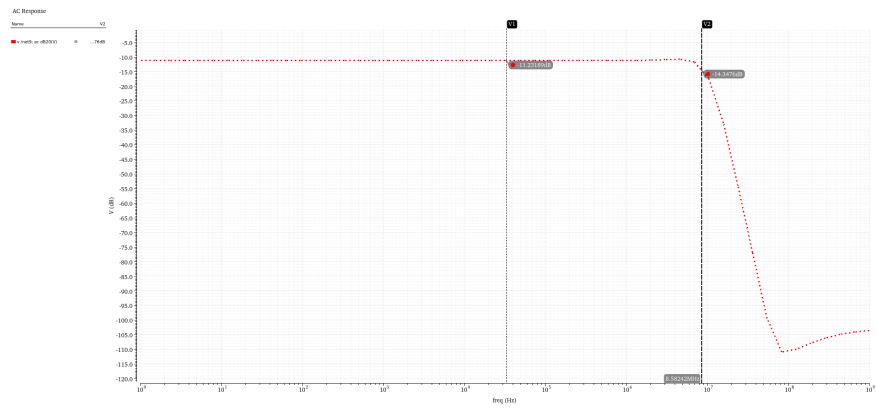


Fig. 29: Post Layout Bode Plot for Gm-cell with Test circuit - Slow Corner

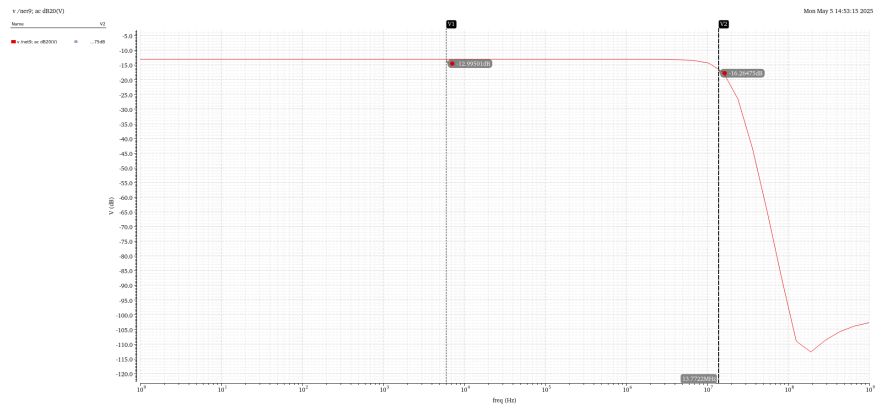


Fig. 30: Post Layout Bode Plot for Gm-cell with Test circuit - Fast corner

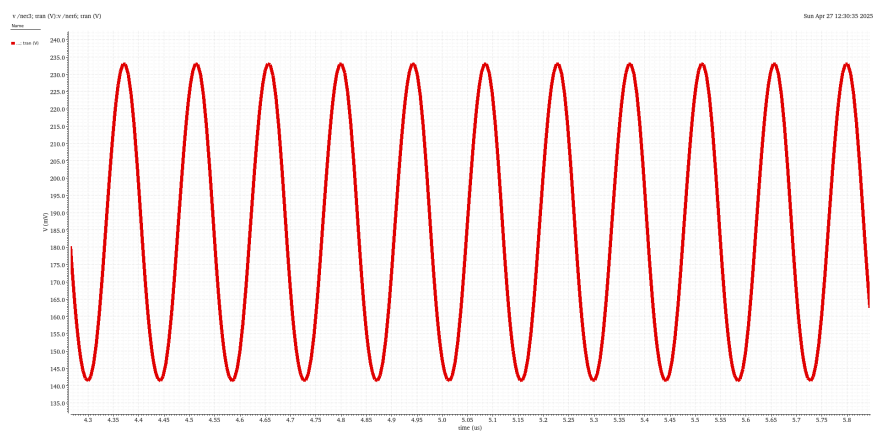


Fig. 31: Post Layout Transient Analysis for Gm-cell with Test circuit

REFERENCES

- 1) Bram Nauta, “A *CMOS Transconductance-C Filter Technique for Very High Frequencies*,” IEEE Journal of Solid-State Circuits.
- 2) Choi Yu Wing, “A *70MHz CMOS Gm-C Bandpass Filter with Automatic Tuning*,” Master of Philosophy Thesis, Department of Electrical and Electronic Engineering, Hong Kong University of Science and Technology.